

E-COMMERCE CUSTOMER PURCHASE PREDICTION

MACHINE LEARNING MINI PROJECT REPORT

Customer Purchase Prediction from E-Commerce

Team Members

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Domain: E-Commerce

Project Type: Binary Classification

Algorithms Used: Logistic Regression, Decision Tree, Random Forest

Target Variable: PurchaseStatus

1. ABSTRACT

This project focuses on predicting whether a customer will make a purchase in an e-commerce environment. The dataset contains 1,500 records and 9 columns, including customer and website-behaviour variables such as age, annual income, number of purchases, product category, time spent on the website, loyalty-program status and discounts availed.

The target variable is PurchaseStatus, where 0 represents no purchase and 1 represents a purchase.

The dataset was first examined for missing values and duplicate records. No missing values were found, while 112 duplicate records were identified and removed, leaving 1,388 records for model development.

Exploratory Data Analysis (EDA) was performed using multiple visualizations to understand customer behaviour and its relationship with purchase status. Three classification algorithms were then implemented: Logistic Regression, Decision Tree and Random Forest.

The models were evaluated using Accuracy, Precision, Recall and F1-score. Random Forest achieved the best overall performance with an accuracy of 92.81% and an F1-score of 92.25%. Therefore, Random Forest was selected as the final model for the project.

2. INTRODUCTION

E-commerce platforms generate a large amount of information related to customer behaviour. Customers differ in their browsing activity, purchasing history, income, website engagement, loyalty-program participation and response to discounts.

Understanding this behaviour can help an e-commerce business identify customers who are more likely to make a purchase.

Machine Learning can be used to identify patterns in historical customer data and make predictions for new customers. In this project, customer purchase prediction is treated as a binary classification problem.

The main aim is not only to train a single Machine Learning model, but to follow a complete Machine Learning workflow including data understanding, data cleaning, preprocessing, exploratory analysis, model training, evaluation, comparison and final prediction.

3. PROBLEM STATEMENT

The problem addressed in this project is to predict whether a customer is likely to make a purchase based on available customer and e-commerce behaviour features.

A successful prediction can help e-commerce businesses better understand customer behaviour and support customer targeting, promotional planning and conversion-related decisions.

Machine Learning is suitable for this problem because the dataset contains multiple input variables along with a known historical outcome. A supervised classification algorithm can learn the relationship between these input variables and the PurchaseStatus target

4. OBJECTIVES

The objectives of the project are:

1. To understand the supplied e-commerce customer dataset.
2. To identify the input features and prediction target.
3. To clean the dataset and remove duplicate records.
4. To check the dataset for missing values and invalid target values.
5. To perform Exploratory Data Analysis using multiple visualizations.
6. To identify patterns in customer behaviour related to purchasing.
7. To train Logistic Regression, Decision Tree and Random Forest classifiers.
8. To evaluate the models using Accuracy, Precision, Recall and F1-score.
9. To compare the performance of the three models.
10. To select the best-performing model.
11. To demonstrate a final purchase prediction using the selected model.
12. To discuss the limitations and future scope of the project.

5. LITERATURE REVIEW

Customer purchase prediction can be treated as a supervised classification problem when the outcome is represented using two classes such as purchase and no purchase.

For customer-related tabular data, different classification approaches can be used to learn relationships between customer characteristics and purchase behaviour.

In this project, three algorithms are compared.

Logistic Regression is used as a baseline classifier because it is simple, fast and suitable for binary classification.

Decision Tree is used because it can represent decision-making through a sequence of feature-based conditions and can capture non-linear patterns.

Random Forest is used as an ensemble method that combines multiple decision trees and can provide more robust predictions.

Comparing these three approaches helps identify which model performs best on the supplied customer dataset.

6. DATASET DESCRIPTION

The project uses the supplied dataset file:

Dataset Name: customer_purchase_data.csv

The dataset contains customer and purchase-related information.

Dataset Summary

Item	Details
Original records	1,500
Columns	9
Input features	8
Target variable	PurchaseStatus
Problem type	Binary Classification
Missing values	0
Duplicate records	112
Records after duplicate removal	1,388

The supplied project files do not record a public external source for the exact CSV file, so the report does not attribute the dataset to an external website without evidence.

Features

1. Age
2. Gender
3. AnnualIncome
4. NumberOfPurchases
5. ProductCategory
6. TimeSpentOnWebsite
7. LoyaltyProgram
8. DiscountsAvailed

Target Variable

PurchaseStatus

- 0 = No Purchase
- 1 = Purchase

7. DATA UNDERSTANDING

The dataset contains 1,500 records and 9 columns before cleaning.

All columns are numeric in the supplied CSV file.

Data Types

Column	Data Type	Role
Age	int64	Feature
Gender	int64	Feature
AnnualIncome	float64	Feature
NumberOfPurchases	int64	Feature
ProductCategory	int64	Feature
TimeSpentOnWebsite	float64	Feature
LoyaltyProgram	int64	Feature

DiscountsAvailable	int64	Feature
PurchaseStatus	int64	Target

The dataset already represents coded categorical or binary variables numerically. Therefore, additional categorical encoding was not required for this supplied dataset.

Target Distribution

After duplicate removal:

- **No Purchase:** 740 records
- **Purchase:** 648 records

The two classes are relatively close in size, although they are not perfectly balanced.

8. DATA PREPROCESSING

Data preprocessing was performed before training the Machine Learning models.

8.1 Missing Value Check

All columns were checked for missing values.

The result showed:

No missing values were present in any column.

Therefore, no missing-value imputation was required.

8.2 Duplicate Removal

A total of **112 duplicate records** were identified.

The dataset changed from:

1,500 records → 1,388 records

after duplicate removal.

8.3 Target Validation

The PurchaseStatus target variable was checked and contained only two values:

- 0
- 1

This confirms that the target is suitable for binary classification.

8.4 Feature and Target Separation

The target variable PurchaseStatus was separated from the input features.

Features (X):

- Age
- Gender
- AnnualIncome
- NumberOfPurchases
- ProductCategory
- TimeSpentOnWebsite
- LoyaltyProgram
- DiscountsAvailed

Target (y):

- PurchaseStatus

8.5 Feature Scaling

Feature scaling was applied for Logistic Regression using StandardScaler.

This was done because the numerical features have different ranges, and scaling helps Logistic Regression work with features on comparable scales.

Decision Tree and Random Forest do not require scaling because their splitting decisions are not dependent on feature magnitude.

8.6 Train-Test Split

The cleaned data was divided into:

- **80% Training Data**
- **20% Testing Data**

The split produced:

- **1,110 training records**
- **278 testing records**

Stratification was used to maintain the target class distribution across the training and testing sets.

9. EXPLORATORY DATA ANALYSIS

Seven visualizations were prepared to understand customer behaviour and its relationship with purchase status.

9.1 Purchase Status Distribution

The first visualization shows the distribution of the target variable.

After duplicate removal:

- 740 customers did not make a purchase.
- 648 customers made a purchase.

Insight

The two groups are reasonably close in size, although they are not perfectly balanced.

9.2 Age Distribution by Purchase Status

This visualization compares the age distribution of purchasing and non-purchasing customers.

Insight

The age distributions of the two groups overlap. Age can contribute to the prediction, but the results suggest that age alone does not determine purchase behaviour.

9.3 Annual Income by Purchase Status

This visualization shows annual income for customers across the two purchase-status groups.

Insight

Annual income varies within both groups. Therefore, income provides useful information, but purchasing behaviour appears to depend on a combination of features rather than income alone.

9.4 Time Spent on Website by Purchase Status

This visualization compares how much time customers spend on the website across the two classes.

Insight

The two groups show different time-spent distributions, indicating that website engagement may contain useful information for predicting purchase intention.

9.5 Loyalty Program Status vs Purchase

This visualization compares customers who are and are not part of the loyalty program.

Insight

Purchase patterns differ between loyalty-program groups, making loyalty-program status a useful feature for the prediction task.

9.6 Correlation Matrix

A correlation heatmap was used to examine the linear relationships between the input features and PurchaseStatus.

The correlations with the target were:

Feature	Correlation with PurchaseStatus
LoyaltyProgram	0.3181
DiscountsAvailable	0.2966
TimeSpentOnWebsite	0.2821
NumberOfPurchases	0.2191
AnnualIncome	0.1912
ProductCategory	0.0143
Gender	0.0089

Age -0.2536

Insight

The strongest positive correlations with PurchaseStatus are observed for LoyaltyProgram, DiscountsAvailed, TimeSpentOnWebsite and NumberOfPurchases. Age shows a negative correlation.

These values indicate association rather than causation.

9.7 Number of Purchases by Purchase Status

This visualization compares customers based on their previous number of purchases.

Insight

The distribution of previous purchase counts differs between the two classes, suggesting that purchase history can help distinguish future purchase outcomes.

10. METHODOLOGY

The project follows the complete Machine Learning workflow:

Problem Identification



Dataset Collection / Loading



Data Understanding



Data Cleaning



Data Preprocessing



Exploratory Data Analysis



Feature Selection



Train-Test Split



Model Training



Model Evaluation



Model Comparison



Best Model Selection



Final Prediction



Conclusion and Future Scope

The same training and testing setup was used for all three algorithms to make the comparison consistent.

11. ALGORITHMS USED

11.1 Logistic Regression

Logistic Regression is a supervised classification algorithm used to predict the probability of a binary outcome.

In this project, it predicts whether:

- `PurchaseStatus = 0` → No Purchase
- `PurchaseStatus = 1` → Purchase

Logistic Regression was selected as the baseline model because it is simple, fast and easy to interpret.

11.2 Decision Tree Classifier

Decision Tree is a classification algorithm that makes predictions through a sequence of feature-based splits.

Each split creates branches based on conditions in the input variables until a final prediction is reached.

In this project, a maximum depth of 5 was used to control the complexity of the tree.

11.3 Random Forest Classifier

Random Forest is an ensemble Machine Learning algorithm that combines the predictions of multiple Decision Trees.

In this project:

- **Number of trees:** 100
- **Maximum depth:** 10

Random Forest was included because it can capture more complex relationships and provide more robust predictions than a single Decision Tree.

12. MODEL TRAINING

The cleaned dataset was divided into training and testing data using an 80:20 split.

The training set contained **1,110 observations**, while the testing set contained **278 observations**.

All three models were trained using the same train-test split:

1. Logistic Regression
2. Decision Tree
3. Random Forest

The models were then used to predict the PurchaseStatus values for the unseen testing data.

13. MODEL EVALUATION

Since the project is a binary classification problem, the following evaluation metrics were used.

13.1 Accuracy

Accuracy represents the proportion of total predictions that were correct.

13.2 Precision

Precision represents the proportion of customers predicted as purchase cases who actually made a purchase.

13.3 Recall

Recall represents the proportion of actual purchase cases that were correctly identified.

13.4 F1-score

F1-score is the harmonic mean of Precision and Recall and provides a balanced measure of classification performance.

13.5 Confusion Matrix

A confusion matrix shows:

- True Positives
- True Negatives
- False Positives
- False Negatives

Confusion matrices were generated for all three models.

14. MODEL COMPARISON

The performance of the three models on the test dataset is shown below.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	83.09%	81.68%	82.31%	81.99%
Decision Tree	85.25%	82.01%	87.69%	84.76%
Random Forest	92.81%	92.97%	91.54%	92.25%

Random Forest achieved the highest performance on the test set. It recorded:

- **Accuracy:** 92.81%
- **Precision:** 92.97%
- **Recall:** 91.54%
- **F1-score:** 92.25%

Decision Tree ranked second, while Logistic Regression produced the lowest scores among the three models.

15. BEST MODEL AND KEY FINDINGS

Best Model: Random Forest Classifier

Random Forest was selected as the final model because it achieved the highest F1-score of **92.25%**.

It also achieved the highest:

- Accuracy
- Precision

among the three models.

Therefore, Random Forest was considered the strongest overall model for the supplied dataset.

Feature Importance

According to the Random Forest model, the most influential features include:

1. TimeSpentOnWebsite
2. Age
3. AnnualIncome
4. DiscountsAvailed
5. NumberOfPurchases
6. LoyaltyProgram
7. ProductCategory
8. Gender

TimeSpentOnWebsite was identified as the most influential feature in the Random Forest model.

These importance values describe how the model uses the features for prediction and should not be interpreted as proof of direct cause and effect.

16. RESULTS AND DISCUSSION

The results show a noticeable difference between the three classification models.

Logistic Regression produced an F1-score of **81.99%**, making it a useful baseline model.

Decision Tree improved the F1-score to **84.76%**, indicating that non-linear decision rules can capture additional patterns in the customer data.

Random Forest achieved the highest F1-score of **92.25%**, indicating that the ensemble of multiple decision trees captured the patterns in the dataset more effectively.

The results suggest that customer purchase behaviour in the supplied dataset is better captured by a tree-based ensemble model than by a single linear decision boundary.

However, the model should be treated as a predictive aid rather than a guarantee of future customer behaviour.

17. FINAL PREDICTION

The selected Random Forest model was also used to demonstrate a prediction for a sample customer.

Sample Customer Input

Feature	Value
Age	30
Gender	1
AnnualIncome	80,000
NumberOfPurchases	10
ProductCategory	2
TimeSpentOnWebsite	35
LoyaltyProgram	1
DiscountsAvailable	3

Prediction

The trained Random Forest model predicted:

Purchase Status: Purchase

Purchase Probability: 97.28%

This example demonstrates how the trained model can be used to estimate whether a customer is likely to make a purchase based on the available customer and website-behaviour features.

18. CONCLUSION

This project demonstrates a complete Machine Learning workflow for customer purchase prediction in the E-Commerce domain.

The dataset was first analysed and cleaned. Missing values were checked and no missing values were found. Duplicate records were removed, reducing the dataset from 1,500 to 1,388 records.

Seven EDA visualizations were prepared to understand customer behaviour and its relationship with purchase status.

Three classification algorithms were implemented and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

Among the three models, Random Forest achieved the best performance with:

- **92.81% Accuracy**
- **92.97% Precision**
- **91.54% Recall**
- **92.25% F1-score**

Therefore, Random Forest was selected as the final model.

The final prediction example also demonstrated how the trained model can be used to estimate whether an individual customer is likely to make a purchase.

19. LIMITATIONS

The project has the following limitations:

1. The dataset contains 1,500 original records and may not represent every type of e-commerce customer or business.
 2. The original external source of the supplied CSV file is not recorded in the project files.
 3. The dataset contains only the features available in the supplied file, so some potentially useful customer or session information may be missing.
 4. Model performance depends on the selected train-test split and model parameters.
 5. The project was evaluated offline and was not tested in a live e-commerce environment.
 6. Feature importance reflects model behaviour and should not be interpreted as direct causal evidence.
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20. FUTURE SCOPE

The project can be improved in the future by:

1. Using a larger and more diverse e-commerce customer dataset.
2. Adding richer customer behaviour and session-level features.
3. Applying hyperparameter tuning to improve model performance.
4. Using cross-validation for more reliable evaluation.
5. Comparing additional algorithms such as Gradient Boosting or Support Vector Machine.
6. Developing a real-time customer purchase prediction service.
7. Using predictions to support personalized offers and customer engagement.
8. Monitoring model performance over time and retraining the model when customer behaviour changes.